

Brishav Debnath

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Education

UPES, Dehradun B.Tech CSE (Hons.) in Big Data	Jul 2023 – Aug 2027 (Expected) CGPA: 9.12/10
LA Garden High School, Ranchi CBSE Higher Secondary Certificate	2020 Percentage: 59%
SHMD School, Jalgaon CBSE Secondary School Certificate	2018 Percentage: 94%

Experience

Malaviya National Institute of Technology (MNIT) Jaipur, Rajasthan Research Intern, Department of Management Studies	May 2026 – Jul 2026 Jaipur, Rajasthan
- Conducted review of more than 90 peer reviewed articles on use of AI for Urban Heat Island (UHI) mitigation through remote sensing, PINNs, fuzzy control, and deep reinforcement learning, writing a review of 150+ pages. - Created a gap matrix for six different capabilities where none of the existing systems enable real time closed loop control of urban cooling. - Created a four layered closed loop architecture (GAN, PINN thermal model, fuzzy safety layer having 150 rules, multi agent Soft Actor-Critic controller) with Energy Return Ratio of ~58. - Wrote a 3-year, INR 1 crore research proposal to DST under 2026 BRICS Call for Proposals.	
Tezpur Central University, Assam Research Intern, Department of Environmental Sciences	Jun 2025 – Jul 2025 Tezpur, Assam
- Applied PCA method to choose 5 low-correlation soil indicators (maximum correlation 0.2) from 23 features and 125 observations and developed a 125-rules fuzzy inference model for soil quality classification. - Cross-validated the knowledge base using RBF kernel SVM for 500 times and improved the accuracy from 82.5% to 88.0% and macro F1 score from 0.799 to 0.823. - Created the Stochastic Frontier Analysis in Python using 720 observations, solved multicollinearity by reducing 40 time dummy variables to 3 and calculated the average technical efficiency of 93.5%.	

Projects

Research Agent - Literature Review Workspace for Obsidian	Jul 2026 – Aug 2026
- Created the architecture and design document for a literature review workspace implemented as a 5-packages monorepo in TypeScript. - Created the design document to integrate the React canvas, a local server, Obsidian vault, and OpenAlex and Semantic Scholar, providing 53 MCP tools and 94 HTTP endpoints using token authentication. - Created the design for an activity log tracking all changes performed by connected AI agents, including conflict resolution and a local LLM router with Ollama.	
Cancer Claim Verifier - Retrieval-Augmented Misinformation Detection	Feb 2026 – Apr 2026
- Trained BioBERT model for classifying health claims into "supported," "contradicted," or "insufficient evidence" categories based on 2,258 claim-evidence pairs to achieve 96.97% accuracy (0.967 macro F1) on 198 test claims and 94.11% ($\pm 1.31\%$) via 30-fold cross-validation. - Identified and removed 64 test claims that were duplicates of the training set (exact/fuzzy match > 90%) prior to evaluation. - Created hybrid retrieval model from a 119-block ChromaDB index, which gives "insufficient evidence" results if there is no relevant source along with displaying with LIME explanations and 31 tests.	
EmotiSense - AI-Powered Mood Tracking System	Mar 2025 – Apr 2025
- Tuned the DistilBERT model using Sentiment140 to achieve 84% accuracy, precision, and recall on sentiment classification with text pre-processing for slang, URLs, mentions, and hashtags. - Designed a Java Swing application to run the Python model and save the data in an SQLite database using JDBC.	
Real-Time E-Commerce Recommendation Pipeline	Apr 2026 – May 2026
- Developed a streaming pipeline employing Kafka and Spark Structured Streaming for calculating trending products and recommending products based on clickstream information, with the use of checkpointing for reliability. - Showed results by providing an interface using FastAPI and MongoDB with Streamlit, deployed via Docker Compose.	

Technical Skills

Programming Languages: Python, Java, SQL, C
Machine Learning and NLP: PyTorch, Hugging Face Transformers (BioBERT, DistilBERT)
Databases and Big Data: Apache Kafka, Apache Spark (Structured Streaming)
Tools and Platforms: Git, GitHub, Docker
Core Concepts: OOP, DSA, DBMS